

The abstract all-order one-dimensional kernel-jet transfer

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Abstract

For $Z(N) = \int_0^b u^h e^{-\beta(Nu^k)^2} F(u, Nu^k) du$ with an amplitude admitting a finite jet expansion $|F(u, s) - \sum_{j < R} u^j F_j(s)| \leq u^R H_R(s)$ at $u = 0$ and Gaussian-weighted moments finite, the density-transfer theorem gives $|Z(N) - \sum_{j < R} c_j N^{-q_j}| \leq K_R N^{-q_R}$ for $N \geq 1$, with $q_j = (h + 1 + j)/k$ and $c_j = k^{-1} \int_0^\infty s^{q_j-1} e^{-\beta s^2} F_j(s) ds$. This is the engine of the all-order theory recommended by the third Astra consult: finite compatible jets with moment-integrable remainders, no analyticity. Zero sorry/axiom.

1 The things formalised

```
noncomputable def oneDKernel ( b N : ) ( h k : ) ( F : → → ) : :=
  u in Icc ( 0 : ) b, u ^ h * (Real.exp ( - * ( N * u ^ k ) ^ 2 ) * F u ( N * u ^ k ))
noncomputable def jetCoeff ( p : ) ( k : ) ( Fj : → → ) ( j : ) : :=
  1 / ( k : ) * logMoment ( p + ( j : ) / k ) 0 ( Fj j )
theorem oneDKernel_jet_expansion ( b p : ) ( h k R : ) ( F : → → ) ( Fj : → → )
  ( HR : → ) ( h : 0 < ) ( hb : 0 < b ) ( hk : 0 < k ) ( hp : (( h : ) + 1 ) / k = p )
  ... ( hrem : u Icc ( 0 : ) b, s, 0 < s →
    | F u s - j Finset.range R, u ^ j * Fj j s | u ^ R * HR s )
(hMc hMt hMH : Gaussian-weighted moment integrability) :
  K : , N : , 1 N →
  | oneDKernel b N h k F
    - j Finset.range R, jetCoeff p k Fj j * N ^ ( -( p + ( j : ) / k )) |
  K * N ^ ( -( p + ( R : ) / k ))
```

2 Proof strategy

The exact density $\rho(r, s) = k^{-1} r^{p-1} F(r^{1/k}, s)$ (unit 85's substitution with a general amplitude) has the **Fin** R -indexed density expansion with exponents $p + j/k$, log degrees 0, coefficients $k^{-1} F_j$ and remainder $r^{p+R/k-1} k^{-1} H_R$, because $r^{p+j/k-1} = r^{p-1} u^j$. The transfer theorem then produces the statement, after converting the transferred terms $N^{-q_j} M_0(k^{-1} F_j)$ and the **Fin**-sums to range sums.

3 Roadblocks and resolutions

- `Integrable.const_mul` of an `IntegrableOn` again needs a type ascription before `.congr_fun`.
- The exact density lemma is stated with $(h + 1)/k$; rewrite it with the exponent hypothesis before using it against the density expansion stated with p .

4 Outcome

The all-order one-dimensional theorem is reduced to producing finite jets with quantitative remainders. Next: the polynomial jet recursion $E_n = \frac{\beta s}{n} \sum_i i x_i E_{n-i}$, $P_j = \sum_\ell y_\ell E_{j-\ell}$ for $\eta e^{\beta s \xi}$ and the jet remainder bound from order- R Taylor remainders of ξ, η , giving $c_j = k^{-1} \sum_\ell p_{j,\ell} S_{q_j+\ell}(a)$.